HealthConnect RAW Layer – Data Model Documentation (2025)

Prepared for: HealthConnect Data Warehouse Project
Version: 1.0 | Environment: RAW Layer | Author: Data Engineering Team

Confidential – For internal use only

Project Overview

The **HealthConnect RAW Layer** stores unprocessed data from various healthcare systems. Each table directly represents source data for auditing, lineage tracking, and integration into the warehouse.

Table: raw_patients

Column	Type	Constraints
patient_id	INT	Primary Key
first_name	VARCHAR(50)	Not Null
last_name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check (M,F,O)
date_of_birth	DATE	Not Null
state_code	CHAR(2)	
city	VARCHAR(50)	
phone	VARCHAR(15)	

Table: raw_providers

Column	Type	Constraints
provider_id	INT	Primary Key
first_name	VARCHAR(50)	Not Null
last_name	VARCHAR(50)	Not Null
specialty	VARCHAR(50)	
npi	VARCHAR(20)	Unique

Table: raw_payers

Column	Type	Constraints
payer_id	INT	Primary Key
payer_name	VARCHAR(100)	Not Null

Table: raw_encounters

Column	Type	Constraints
encounter_id	INT	Primary Key
patient_id	INT	Not Null (FK → raw_patients)
provider_id	INT	Not Null (FK → raw_providers)
encounter_type	VARCHAR(10)	Check (OPD, ER, IPD)
encounter_start	DATETIME	
encounter_end	DATETIME	
height_cm	INT	
weight_kg	FLOAT	
systolic_bp	INT	
diastolic_bp	INT	

Table: raw_diagnoses

Column	Type	Constraints
diagnosis_id	INT	Primary Key
encounter_id	INT	Not Null (FK → raw_encounters)
diagnosis_description	VARCHAR(100)	
is_primary	BIT	

Table: raw_procedures

Column	Type	Constraints
procedure_id	INT	Primary Key
encounter_id	INT	Not Null (FK → raw_encounters)
procedure_description	VARCHAR(100)	

Table: raw_medications

Column	Type	Constraints
medication_id	INT	Primary Key
encounter_id	INT	Not Null (FK → raw_encounters)
drug_name	VARCHAR(50)	
route	VARCHAR(20)	
dose	VARCHAR(20)	
frequency	VARCHAR(10)	
days_supply	INT	

Table: raw_claims

Column	Type	Constraints
claim_id	INT	Primary Key
encounter_id	INT	Not Null (FK → raw_encounters)
payer_id	INT	Not Null (FK → raw_payers)
admit_date	DATE	
discharge_date	DATE	
total_billed_amount	FLOAT	
total_allowed_amount	FLOAT	
total_paid_amount	FLOAT	
claim_status	VARCHAR(20)	Check (Pending, Approved, Rejected, Paid)